

Lab 3: To study the diode applications in clipping and clamping circuits.

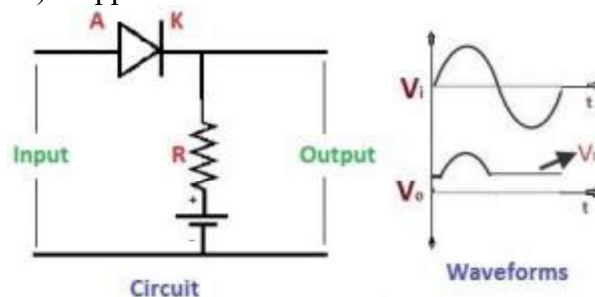
Objective:

- To observe the waveforms of the various types (Positive and Negative) of clipper circuits.
- To observe the waveforms of the Positive and Negative clamping circuits.

Clipper circuits:

Clipper circuits have the ability to “clip” off a portion of the input signal without distorting the remaining part of the alternating waveform. There are two general categories of clippers: series and parallel. The series configuration is defined as one where the diode is in series with the load, while the parallel variety has the diode in branch parallel to the load. Clippers are useful for protecting circuits from exceeding various voltages (either positive or negative).

A. Series Biased (Negative) Clipper



Series Negative Clipper with positive biasing voltage

Series negative clipper with positive reference voltage is similar to the series negative clipper, but in this a positive reference voltage is added in series with the resistor. During the positive half cycle, the diode start conducting only after its anode voltage value exceeds the cathode voltage value. Since cathode voltage becomes equal to the reference voltage, the output that appears across the resistor will be as shown in the above figure.

Circuit Diagram for simulation:

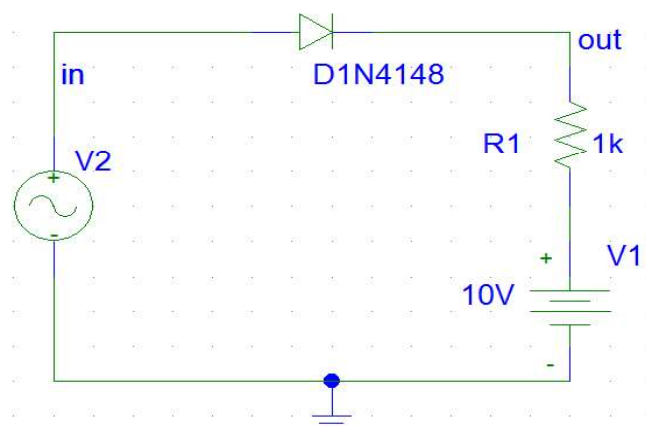
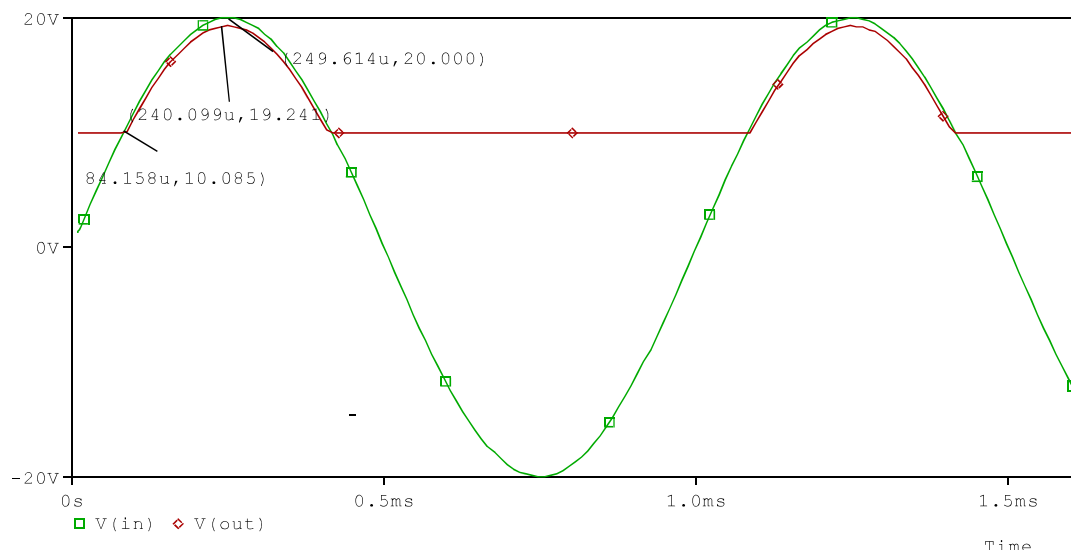
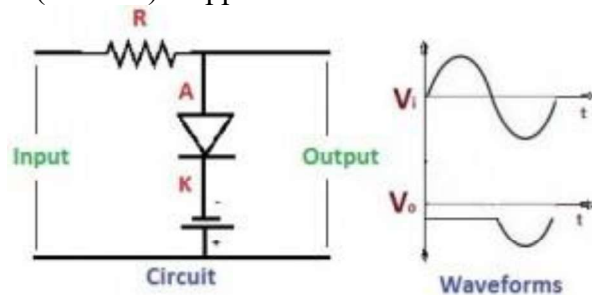


Figure 01

Simulation Result:



B. Parallel (Shunt) Biased (Positive) Clipper



Shunt Positive Clipper with negative biasing voltage

During the positive half cycle, the negative reference voltage connected in series with the diode appears as output; and during the negative half cycle, the diode conducts until the input voltage value becomes greater than the negative reference voltage and output will be generated as shown in the figure.

Circuit diagram for simulation:

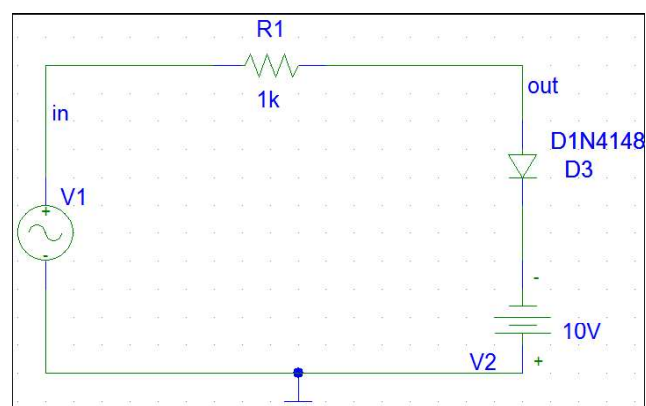
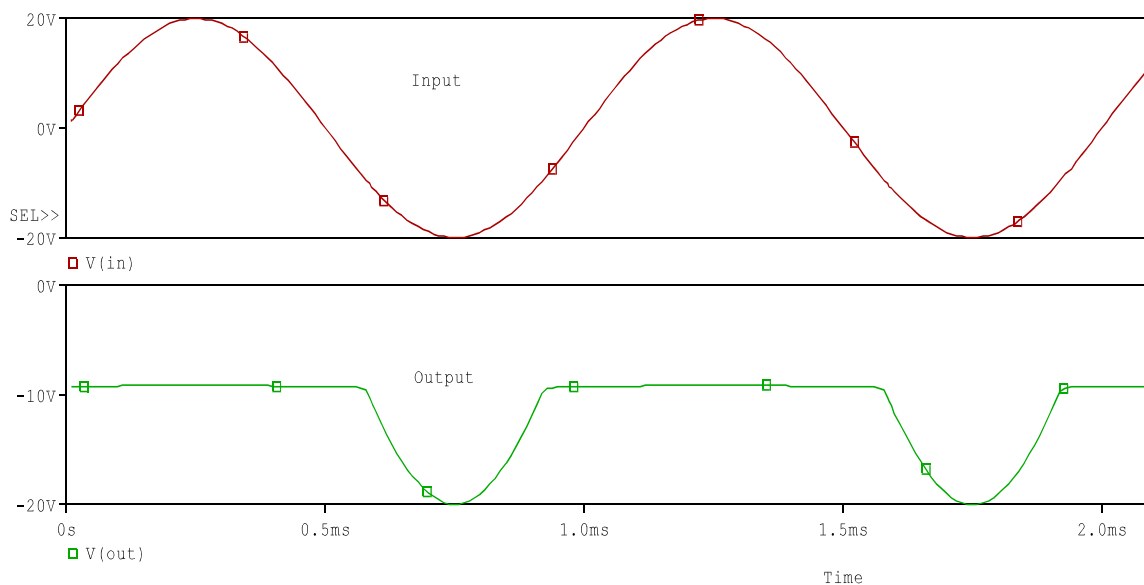


Figure 02

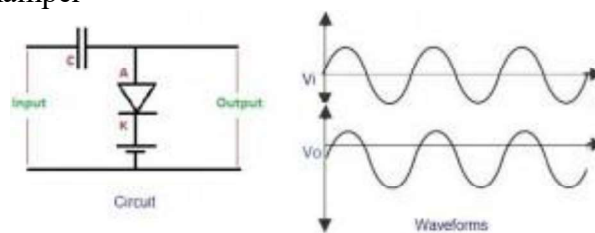
Simulation result:



Clamper Circuits:

The positive or negative peak of a signal can be positioned at the desired level by using the clamping circuits. As we can shift the levels of peaks of the signal by using a clamper, hence, it is also called as level shifter. The clamper circuit consists of a capacitor and diode connected in parallel across the load. The clamper circuit depends on the change in the time constant of the capacitor. The capacitor must be chosen such that, during the conduction of the diode, the capacitor must be sufficient to charge quickly and during the non-conducting period of diode, the capacitor should not discharge drastically. The clammers are classified as positive and negative clammers based on the clamping method.

C. Negative (Biased) Clamper



Negative Clamper with positive basing voltage

It is similar to the negative clamper, but the output waveform is shifted towards the positive direction by a positive reference voltage. As the positive reference voltage is connected in series with the diode, during the positive half cycle, even though the diode conducts, the output voltage becomes equal to the reference voltage; hence, the output is clamped towards the positive direction as shown in the above figure.

Circuit diagram for simulation:

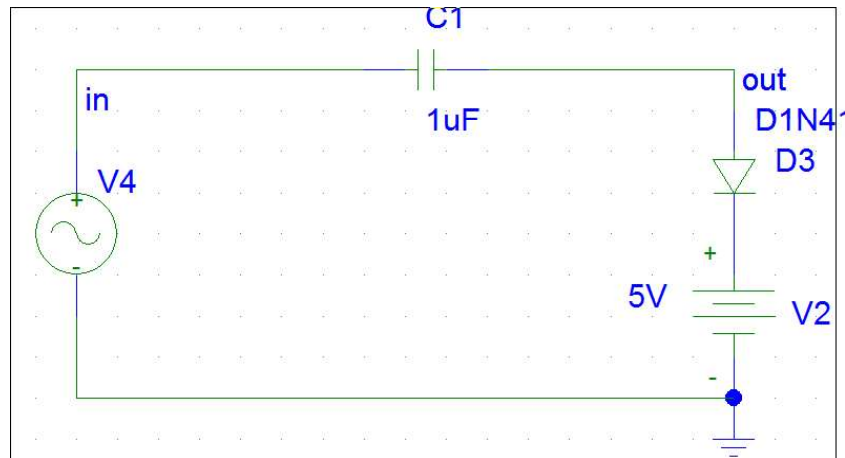
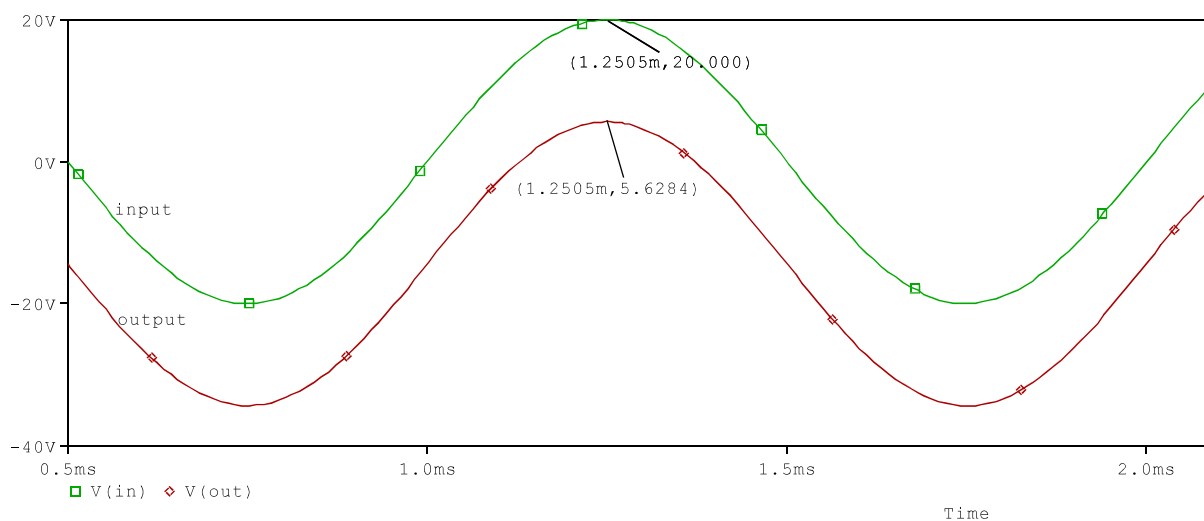
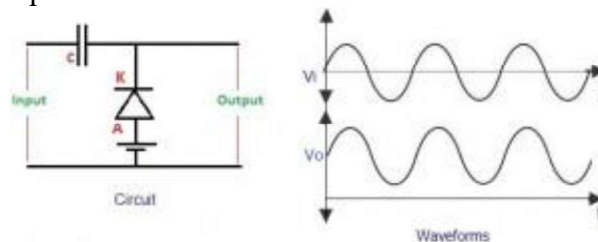


Figure 03

Simulation result:



D. Positive (Biased) Clamper



Positive Clamper with positive biasing voltage

A positive reference voltage is added in series with the diode of the positive clamper as shown in the circuit. During the positive half cycle of the input, the diode conducts as initially the supply voltage is less than the anode positive reference voltage. If once the cathode voltage is greater

than anode voltage then the diode stops conduction. During the negative half cycle, the diode conducts and charges the capacitor. The output is generated as shown in the figure.

Circuit diagram for simulation:

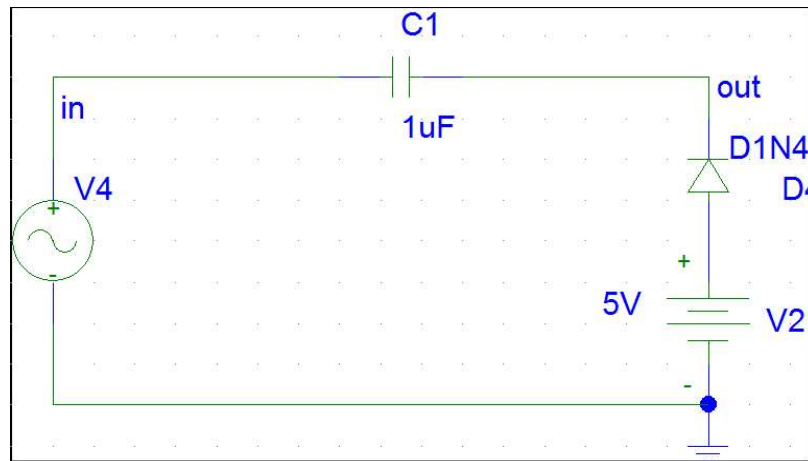
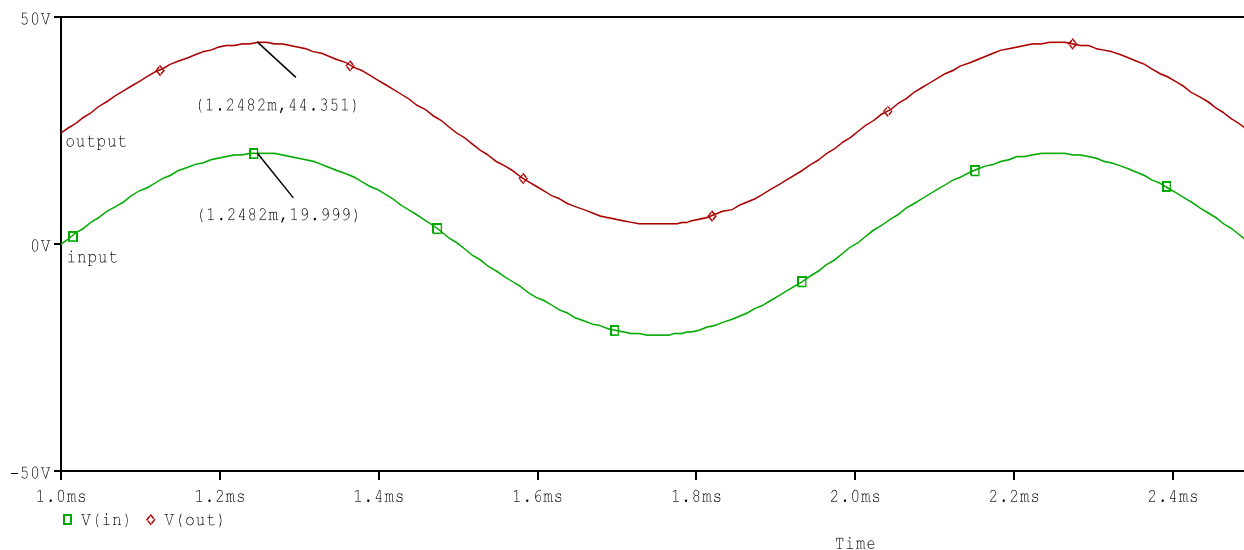


Figure 04

Simulation result:



Post lab report questions:

1. If the diodes in fig.1 and fig.2 are connected in opposite/reverse direction, then how the outputs will change – show in simulation diagram. [Also, provide the modified circuit diagrams with your simulation results]
2. If the fixed +5V biasing source in fig.3 and fig.4 are replaced with a fixed -5V biasing source, then how the outputs will change – show in simulation diagram. [Also, provide the modified circuit diagrams with your simulation results]